

# Experiment 5

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## 1 Aim

To perform  $\beta$  ray spectroscopy of  $^{22}_{11}\text{Na}$  and  $^{90}_{38}\text{Sr}$ . And then analyse the kinetic energy distribution of  $\beta$  particle emitted, with increasing magnetic field.

## 2 Experimental setup



**Figure 1:** Setup of the instruments at the laboratory

First we insert the source and the Hall probe inside the spectrometer. Before beginning with readings of each of the sources, the teslameter was adjusted for the null offset (discussed later) due to 0 current. In both the sources, the counts were taken by increasing the coil voltage ( $V$ ) whilst going to a maximum of coil current ( $I$ ) of 2 A. Further increase in current was prohibited for the safety of the instruments. Multiple readings were taken for each count to minimize statistical errors. The time  $t$  for each source's readings was adjusted keeping in mind the time availability. The directions of the magnetic field were reversed by interchanging the direction of current. The energy of  $\beta$  particle  $E_\beta$  was then calculated for

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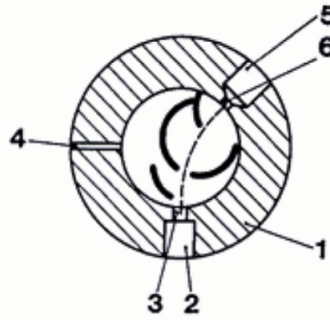
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each magnetic field ( $B$ ) using the kinetic energy equation derived in the later section, and a graph was plotted showing the relationship between the counts per 10 sec ( $\frac{N}{10\text{ s}}$ ) and  $E_\beta$ .

### 3 Working Principles

#### 3.1 $\beta$ - ray spectrometer

The  $\beta$ -ray spectroscope operates by selecting  $\beta$  radiation emitted from unstable atomic nuclei based on the particles' momentum as they pass through a transverse magnetic field. This selection occurs through a diaphragm system that directs beta particles into a predetermined path within a homogeneous magnetic field. In the field, each beta particle experiences a Lorentz force, which balances with the centrifugal force acting on it, resulting in a stable circular orbit. The relationship between the particle's momentum ( $p$ ), charge ( $e$ ), magnetic flux density ( $B$ ), and orbital radius ( $r$ ) is defined by  $p = eBr$ . The radius of this orbit depends on the particle's velocity, charge, and the magnetic field strength, which together determine the particle's momentum. A unique particle energy corresponds to each magnetic field strength, allowing the device to filter particles by energy. For calibration, the spectrometer measures the correlation between coil current, which determines the magnetic field strength, and particle energy, making it possible to quantify the energy distribution of beta particles in the beta spectrum.



**Figure 2:** The components are: 1) non-magnetisable wall; 2) specimen entrance; 3) iris; 4) entrance for tangential Hall probe; 5) counter tube holder; 6) iris.

#### 3.2 Geiger Müller (GM) Counter

These detectors work by using an electric field to collect and count ion pairs (electrons and positive ions) created when radiation ionizes a gas inside a sealed chamber. The GM counter consists of a cylindrical metallic chamber that serves as the cathode, with a fine wire along its axis as the anode (depicted in figure below). Inert(noble) gases are chosen to fill the counter for their low electron affinity - they do not tend to form negative ions. Argon is most common fill gas: It is inert, stable (non-radioactive), abundant, inexpensive, has reasonably high atomic number and works well with halogen quenching gases (See Fig 3).

In a cylindrical geometry, electric field( $\mathcal{E}$ ) at a radial distance  $r$  from the center is given by

$$\mathcal{E}(r) = \frac{V}{r \ln\left(\frac{b}{a}\right)}$$

where  $V$  is voltage applied between anode and cathode,  $a$  is anode wire radius,  $b$  is cathode inner radius.

This shows that the electric field strength for a given voltage is much larger in cylindrical geometry than in a parallel plate one, which makes the experimental conditions easier to attain. Another advantage of the cylindrical geometry is that *avalanche* (amplification) takes place only in narrow region close to the anode wire which gives good uniformity.

#### Townsend Avalanche

The chamber has a sealed window through which radiation like alpha ( $\alpha$ ), beta ( $\beta$ ), or gamma ( $\gamma$ ) particles enter and ionize the gas, typically argon at low pressure. Usually monatomic gas is preferred

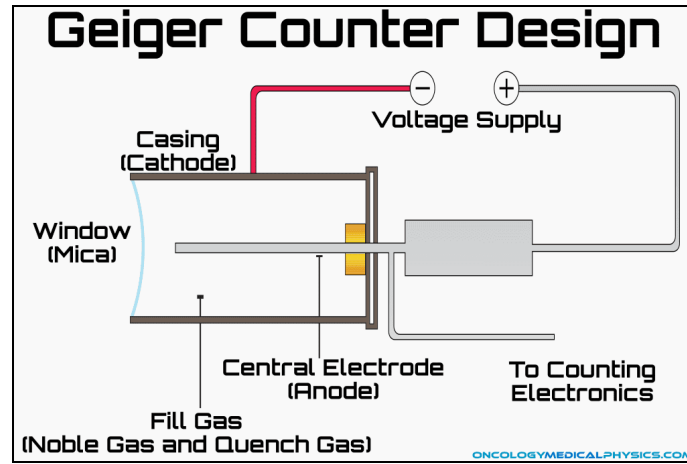


Figure 3: Schematic of GM counter

so that deposited energy is not dissipated in dissociating the molecules. This ionization produces free electrons and positive ions. A high voltage applied across the chamber generates an electric field that accelerates these free electrons toward the anode and the positive ions toward the cathode. This movement of ions creates a current pulse as they reach their respective electrodes. To amplify the initial ionization, the GM counter operates at a high voltage (often over 1000 V), allowing the electrons to gain enough energy to cause secondary ionisations as they collide with other gas atoms in a process called a *Townsend Avalanche*. This avalanche effect exponentially increases the number of ions and electrons, resulting in a strong pulse of current that the counter registers (diagram below).

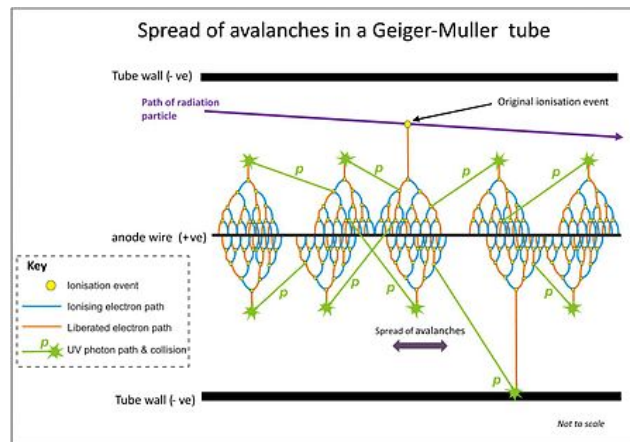
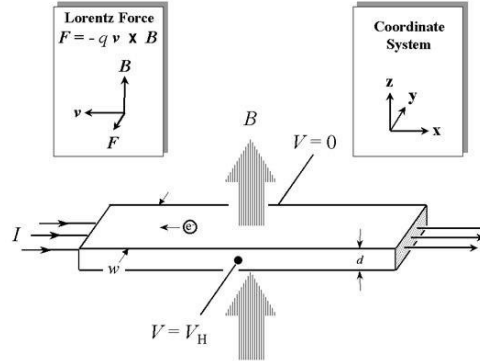


Figure 4: Townsend Avalanche

However, without control, the GM tube might experience continuous discharge, making it unresponsive to new ionising events. To prevent this, a *quenching agent*, such as butane, propane, or halogen gases (like chlorine or bromine), is added. The quenching agent absorbs excess energy and neutralises positive ions without triggering further ionization. In organic quenching gases, energy from the positively charged ions is dissipated by breaking molecular bonds, which stops continuous discharge. Once the ions reach the electrodes and are neutralised, the tube resets, ready to detect the next radiation event. If continuous ionization is not prevented the detector will not respond to any other particle except the very first which enters the tube. The GM counter's design makes it suitable for measuring low to moderate levels of radiation, though it cannot provide information about the energy of incoming particles since all pulse sizes are identical regardless of the intensity of the radiation. Due to its simplicity and robustness, the GM counter is popular as a portable radiation detector, though it is limited to applications requiring simple presence detection rather than detailed energy spectroscopy.

### 3.3 Teslameter

A Teslameter with a Hall probe is an instrument used to measure magnetic field strength. It operates on the principle of the **Hall effect**, which states that when a conductor or semiconductor carrying current is placed in a perpendicular magnetic field, a transverse electric field is generated, resulting in a measurable voltage. This voltage is known as the **Hall voltage**.



#### Hall Effect in the Probe

When current flows through a thin, flat Hall probe in the presence of a perpendicular magnetic field  $\vec{B}$ , Lorentz force causes charge carriers (typically electrons or holes) to deflect to one side of the probe. This deflection creates a voltage difference across the sides of the probe, called the Hall voltage ( $V_H$ ), given by:

$$V_H = \frac{IB}{nqd}$$

where  $I$  = current flowing through the probe,  $B$  = magnetic field strength,  $n$  = charge carrier density in the material,  $q$  = charge magnitude of the carrier (e.g.,  $e$  for electrons),  $d$  = thickness of the Hall probe.

#### Measurement of Magnetic Field

The Hall voltage  $V_H$  is directly proportional to the magnetic field strength  $B$  (for a given current  $I$  and material properties). The teslameter reads this voltage and, based on the known characteristics of the Hall probe, converts it into a magnetic field measurement. Thus, the magnetic field  $B$  can be expressed as:

$$B = \frac{V_H \cdot n \cdot q \cdot t}{I}$$

To ensure accurate readings, the Hall probe is calibrated against known magnetic field strengths. The Teslameter's internal electronics are pre-calibrated to convert the Hall voltage into a direct reading of magnetic field strength, typically displayed in units of Tesla (T) or Gauss (G). Hall probes are sensitive to the direction of the magnetic field. Because they are compact and can measure both DC and AC magnetic fields, they are widely used in laboratories and industrial settings.

## 4 Theory

Beta decay is a type of radioactive decay where an unstable atomic nucleus transitions to a more stable state by emitting beta particles. This transformation can occur primarily through two modes: beta minus decay ( $\beta^-$ ) and beta plus decay ( $\beta^+$ ). Additionally, electron capture is another process governed by the weak nuclear force. The theoretical basis of these decay processes lies in fundamental particle interactions.

### $\beta^-$ Decay

In beta minus decay, a neutron within the nucleus changes into a proton, releasing a beta particle (electron) and an electron antineutrino. This reaction is expressed as:

$$n \rightarrow p + e^- + \bar{\nu}_e$$

### $\beta^+$ Decay

In beta plus decay, a proton transforms into a neutron, releasing a positron (the antimatter version of an electron) and a neutrino:

$$p \rightarrow n + e^+ + \nu_e$$

### Electron Capture

Electron capture is a process where the nucleus captures one of its inner electrons, resulting in the conversion of a proton into a neutron. The reaction can be represented as:

$$p + e^- \rightarrow n + \nu_e$$

### Q-value

The energy released in beta decay and electron capture is determined by the Q value, which reflects the difference in binding energies between the parent and daughter nuclei. The Q value is calculated using the mass-energy equivalence  $E = mc^2$ :

$$Q = (m_{\text{parent}} - m_{\text{daughter}} - m_e)c^2$$

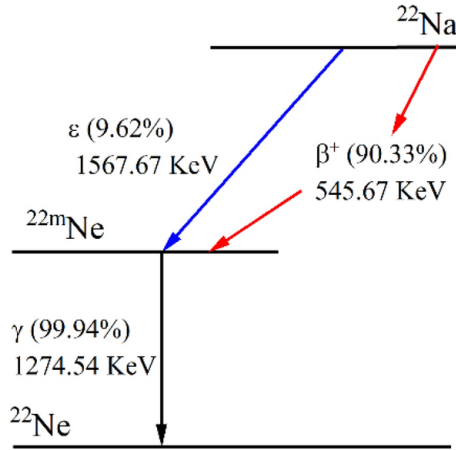
Where  $m_{\text{parent}}$  is the mass of the original nucleus,  $m_{\text{daughter}}$  is the mass of the nucleus after decay,  $m_e$  is the mass of the emitted electron or positron (or negligible for electron capture).



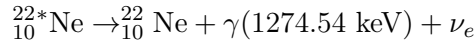
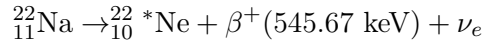
**Figure 5:** The sources used in our experiment

DECAY SCHEME OF  $^{22}_{11}\text{Na}$ 

This radio nuclide undergoes a  $\beta^+$  decay with the emission of a positron and a neutrino.



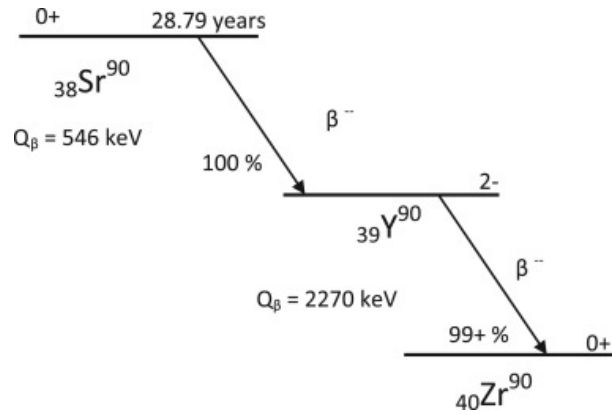
The detailed schematic is as follows:



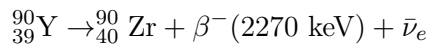
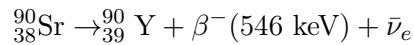
In the first decay process has  $t_{1/2} \approx 2.6$  yrs; while the de-excitation of the nucleus is almost immediate ( $\sim 1$  pico-second).

 DECAY SCHEME OF  $^{90}_{38}\text{Sr}$ 

This radio nuclide undergoes a  $\beta^-$  decay with the emission of an electron and an antineutrino.

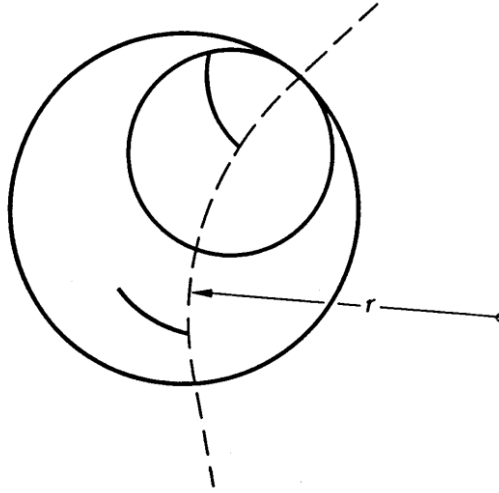


The detailed schematic is as follows:



In the first decay process has  $t_{1/2} \approx 28.8$  yrs, while the later has a comparatively shorter  $t_{1/2} \approx 64$  hrs.

# $\beta$ PARTICLE KINETIC ENERGY



**Figure 6:** Diaphragm system and a magnetically deflected  $\beta$ -particle.  $r$  is the orbital radius.

The  $\beta$ -particles are selected in the  $\beta$ -spectroscopie on the basis of their energy by forcing them to follow a fixed orbit using diaphragms (see Fig 6) in a homogeneous magnetic field ( $\mathbf{B}\hat{z}$ ). In this orbit the Lorentz force balances the centrifugal force for the particle to be in equilibrium. Let  $q$  be the charge of the  $\beta$  particle ( $q = +e$  for  $\beta^+$  ;  $q = -e$  for  $\beta^-$ ).

$$\begin{aligned}\mathbf{F}_B &= \mathbf{F}_{cen} \\ qvB &= \frac{mv^2}{r} = \frac{pv}{r} \\ \implies p &= qBr\end{aligned}$$

Now from special theory of relativity, we know that the total energy of a particle ( $E$ ) can be expressed as the sum of its kinetic energy ( $E_{kin}$ ) and rest mass ( $m_0$ ) energy.

$$E = \gamma m_0 c^2 = E_{kin} + m_0 c^2$$

where,  $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$ . Also we have the famous *energy-momentum* relationship for relativistic particles

$$E = \sqrt{m_0^2 c^4 + p^2 c^2}$$

plugging it in the above expressions, we obtain

$$E_{kin} = c\sqrt{m_0^2 c^2 + (qBr)^2} - m_0 c^2$$

In our experiment we have

|       |                              |
|-------|------------------------------|
| $r$   | 50 mm                        |
| $c$   | $299792.5 \times 10^3$ m/s   |
| $m_0$ | $9.10191 \times 10^{-31}$ kg |
| $e$   | $1.60210 \times 10^{-19}$ C  |

## 5 Data Analysis

We have performed this experiment with two different sources, namely  $^{90}_{38}\text{Sr}$  and  $^{22}_{11}\text{Na}$ . The kinetic energy for the  $\beta$  particle in both the cases has been calculated using the formula ( $|q| = e$ ) we derived earlier.

$$E_{\text{kin}} = c\sqrt{m_0^2c^2 + (eBr)^2} - m_0c^2$$

where the quantities are already defined in the preceding section. Note that, we must need to reverse the direction of the magnetic field because the emitted  $\beta$  particles in these radio-nuclides are of opposite charges.

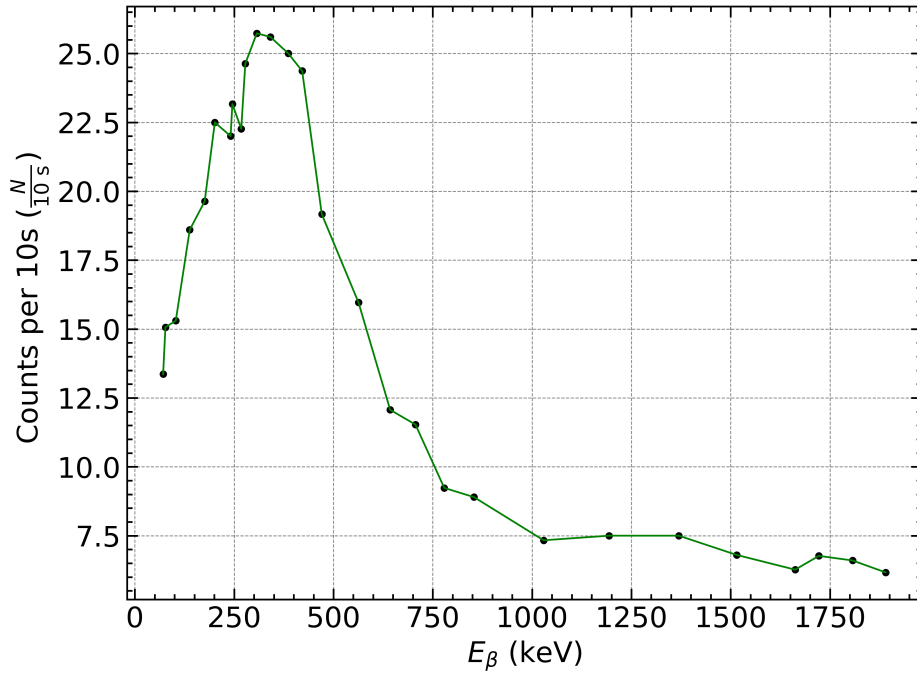
### $\beta^+$ SPECTRUM OF $^{22}_{11}\text{Na}$

We take 3 readings for  $t = 100\text{s}$  and took their average to minimize statistical errors. Before beginning our experiment (i.e at  $I = 0$ ), we adjusted for the the null offset of the tesla-meter (i.e magnetic field  $B = 0.0\text{ mT}$  at  $I = 0.00\text{ A}$ ).

**Table 1:** Readings for  $^{22}_{11}\text{Na}$

| Sl. no. | V (V) | I (A) | B (mT)  | Energy ( $E_\beta$ ) (keV) | Count 1 | Count 2 | Count 3 | N      | $\frac{N}{100\text{ s}} \times 10\text{ s}$ | $\sigma$ |
|---------|-------|-------|---------|----------------------------|---------|---------|---------|--------|---------------------------------------------|----------|
| 1       | 0.11  | 0.04  | -18.60  | 71.1625                    | 142     | 127     | 132     | 133.67 | 13.37                                       | 7.6376   |
| 2       | 0.25  | 0.07  | -19.40  | 77.0039                    | 147     | 150     | 155     | 150.67 | 15.07                                       | 4.0415   |
| 3       | 0.31  | 0.13  | -22.70  | 102.992                    | 150     | 153     | 156     | 153.00 | 15.30                                       | 3.0000   |
| 4       | 0.41  | 0.19  | -26.70  | 138.164                    | 199     | 173     | 186     | 186.00 | 18.60                                       | 13.0000  |
| 5       | 0.61  | 0.25  | -30.60  | 175.773                    | 207     | 189     | 193     | 196.33 | 19.63                                       | 9.4516   |
| 6       | 0.72  | 0.30  | -33.10  | 201.363                    | 225     | 220     | 230     | 225.00 | 22.50                                       | 5.0000   |
| 7       | 0.81  | 0.34  | -36.80  | 241.068                    | 243     | 217     | 200     | 220.00 | 22.00                                       | 21.6564  |
| 8       | 0.86  | 0.35  | -37.20  | 245.479                    | 240     | 230     | 225     | 231.67 | 23.17                                       | 7.6376   |
| 9       | 0.91  | 0.36  | -39.20  | 267.853                    | 216     | 232     | 220     | 222.67 | 22.27                                       | 8.3267   |
| 10      | 0.95  | 0.38  | -40.10  | 278.086                    | 233     | 251     | 255     | 246.33 | 24.63                                       | 11.7189  |
| 11      | 1.01  | 0.42  | -42.60  | 307.007                    | 261     | 243     | 268     | 257.33 | 25.73                                       | 12.8970  |
| 12      | 1.10  | 0.46  | -45.50  | 341.391                    | 260     | 245     | 263     | 256.00 | 25.60                                       | 9.6437   |
| 13      | 1.21  | 0.50  | -49.20  | 386.407                    | 254     | 252     | 244     | 250.00 | 25.00                                       | 5.2915   |
| 14      | 1.30  | 0.54  | -52.00  | 421.222                    | 257     | 224     | 250     | 243.67 | 24.37                                       | 17.3877  |
| 15      | 1.42  | 0.59  | -55.90  | 470.648                    | 194     | 178     | 203     | 191.67 | 19.17                                       | 12.6623  |
| 16      | 1.61  | 0.66  | -63.00  | 562.962                    | 174     | 154     | 151     | 159.67 | 15.97                                       | 12.5033  |
| 17      | 1.81  | 0.75  | -69.00  | 642.871                    | 112     | 131     | 119     | 120.67 | 12.07                                       | 9.6090   |
| 18      | 1.90  | 0.77  | -73.70  | 706.445                    | 110     | 120     | 116     | 115.33 | 11.53                                       | 5.0332   |
| 19      | 2.10  | 0.86  | -79.00  | 778.993                    | 93      | 77      | 107     | 92.33  | 9.23                                        | 15.0111  |
| 20      | 2.32  | 0.95  | -84.40  | 853.701                    | 72      | 86      | 109     | 89.00  | 8.90                                        | 18.6815  |
| 21      | 2.72  | 1.11  | -96.90  | 1029.06                    | 75      | 69      | 76      | 73.33  | 7.33                                        | 3.7859   |
| 22      | 3.20  | 1.25  | -108.50 | 1194.08                    | 73      | 77      | 75      | 75.00  | 7.50                                        | 2.0000   |
| 23      | 3.51  | 1.42  | -120.70 | 1369.36                    | 78      | 76      | 71      | 75.00  | 7.50                                        | 3.6056   |
| 24      | 3.90  | 1.53  | -130.80 | 1515.48                    | 71      | 66      | 67      | 68.00  | 6.80                                        | 2.6458   |
| 25      | 4.32  | 1.68  | -140.90 | 1662.33                    | 57      | 68      | 63      | 62.67  | 6.27                                        | 5.5076   |
| 26      | 4.60  | 1.71  | -145.00 | 1722.12                    | 68      | 65      | 70      | 67.67  | 6.77                                        | 2.5166   |
| 27      | 4.72  | 1.83  | -150.80 | 1806.84                    | 70      | 59      | 69      | 66.00  | 6.60                                        | 6.0828   |
| 28      | 5.02  | 1.93  | -156.50 | 1890.26                    | 61      | 67      | 57      | 61.67  | 6.17                                        | 5.0332   |





**Figure 7:** Plot of average counts per 10s with varying energy of  $\beta^+$  particles for Na-22 decay.

### $\beta^-$ SPECTRUM OF $^{90}_{38}\text{Sr}$

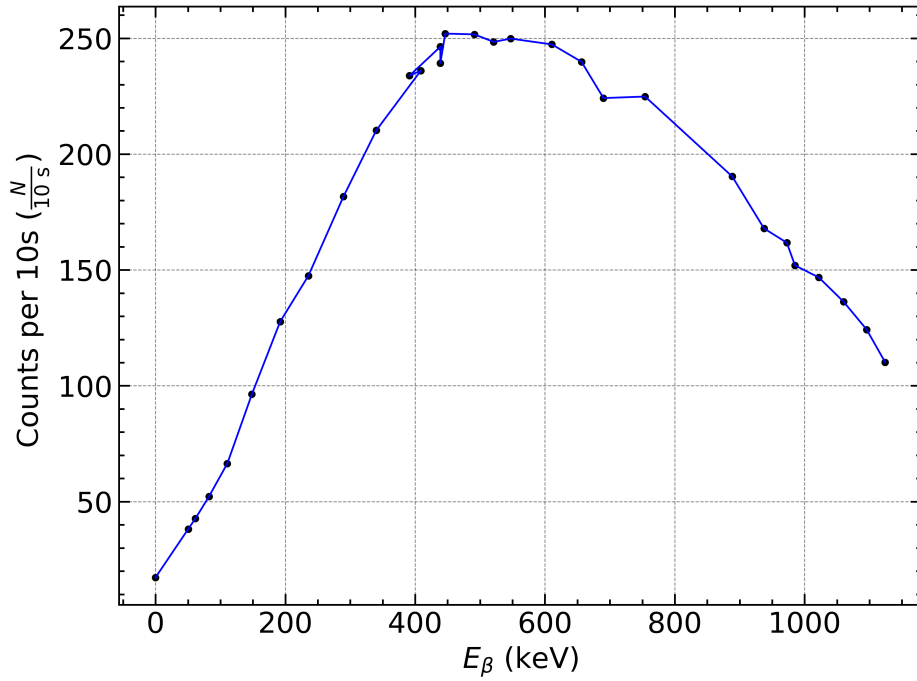
Here we take 2 readings for  $t = 60s$ .

**Table 2:** Readings for  $^{90}_{38}\text{Sr}$

| Sl. no. | V (V) | I (A) | B (mT) | Energy ( $E_\beta$ ) (keV) | Count 1 | Count 2 | N       | $\frac{N}{60s} \times 10s$ | $\sigma$ |
|---------|-------|-------|--------|----------------------------|---------|---------|---------|----------------------------|----------|
| 1       | 0.00  | 0.00  | 0.0    | 0.0                        | 119     | 88      | 103.50  | 17.2500                    | 21.9203  |
| 2       | 0.15  | 0.06  | 15.5   | 50.4                       | 234     | 224     | 229.00  | 38.1667                    | 7.0711   |
| 3       | 0.25  | 0.10  | 17.2   | 61.4                       | 242     | 270     | 256.00  | 42.6667                    | 19.7990  |
| 4       | 0.44  | 0.19  | 20.1   | 82.3                       | 303     | 323     | 313.00  | 52.1667                    | 14.1421  |
| 5       | 0.64  | 0.27  | 23.6   | 111                        | 377     | 420     | 398.50  | 66.4167                    | 30.4056  |
| 6       | 0.85  | 0.36  | 27.8   | 148                        | 552     | 604     | 578.00  | 96.3333                    | 36.7696  |
| 7       | 1.05  | 0.45  | 32.2   | 192                        | 766     | 766     | 766.00  | 127.6667                   | 0.0000   |
| 8       | 1.25  | 0.53  | 36.3   | 236                        | 885     | 885     | 885.00  | 147.5000                   | 0.0000   |
| 9       | 1.45  | 0.61  | 41.1   | 290                        | 1076    | 1104    | 1090.00 | 181.6667                   | 19.7990  |
| 10      | 1.65  | 0.68  | 45.4   | 340                        | 1273    | 1250    | 1261.50 | 210.2500                   | 16.2635  |
| 11      | 1.75  | 0.70  | 51.0   | 409                        | 1434    | 1398    | 1416.00 | 236.0000                   | 25.4558  |
| 12      | 1.85  | 0.77  | 49.6   | 391                        | 1426    | 1380    | 1403.00 | 233.8333                   | 32.5269  |
| 13      | 1.96  | 0.80  | 53.4   | 439                        | 1393    | 1563    | 1478.00 | 246.3333                   | 120.2082 |
| 14      | 2.05  | 0.87  | 53.4   | 439                        | 1474    | 1397    | 1435.50 | 239.2500                   | 54.4472  |
| 15      | 2.16  | 0.91  | 54.0   | 446                        | 1513    | 1511    | 1512.00 | 252.0000                   | 1.4142   |
| 16      | 2.25  | 0.95  | 57.5   | 491                        | 1483    | 1537    | 1510.00 | 251.6667                   | 38.1838  |
| 17      | 2.36  | 0.96  | 59.8   | 521                        | 1542    | 1439    | 1490.50 | 248.4167                   | 72.8320  |
| 18      | 2.45  | 1.03  | 61.8   | 547                        | 1530    | 1468    | 1499.00 | 249.8333                   | 43.8406  |
| 19      | 2.64  | 1.12  | 66.6   | 611                        | 1485    | 1483    | 1484.00 | 247.3333                   | 1.4142   |
| 20      | 2.85  | 1.20  | 70.0   | 656                        | 1474    | 1404    | 1439.00 | 239.8333                   | 49.4975  |
| 21      | 3.05  | 1.28  | 72.5   | 690                        | 1333    | 1357    | 1345.00 | 224.1667                   | 16.9706  |
| 22      | 3.25  | 1.36  | 77.2   | 754                        | 1372    | 1326    | 1349.00 | 224.8333                   | 32.5269  |

*Continued on next page*

| Sl. no. | V (V) | I (A) | B (mT) | Energy ( $E_\beta$ ) (keV) | Count 1 | Count 2 | N       | $\frac{N}{60\text{ s}} \times 10\text{ s}$ | $\sigma$ |
|---------|-------|-------|--------|----------------------------|---------|---------|---------|--------------------------------------------|----------|
| 23      | 3.46  | 1.43  | 86.9   | 889                        | 1132    | 1152    | 1142.00 | 190.3333                                   | 14.1421  |
| 24      | 3.65  | 1.52  | 90.4   | 937                        | 1024    | 991     | 1007.50 | 167.9167                                   | 23.3345  |
| 25      | 3.84  | 1.60  | 92.9   | 973                        | 975     | 966     | 970.50  | 161.7500                                   | 6.3640   |
| 26      | 4.05  | 1.67  | 93.8   | 985                        | 902     | 921     | 911.50  | 151.9167                                   | 13.4350  |
| 27      | 4.25  | 1.75  | 96.4   | 1020                       | 893     | 868     | 880.50  | 146.7500                                   | 17.6777  |
| 28      | 4.45  | 1.82  | 99.1   | 1060                       | 803     | 832     | 817.50  | 136.2500                                   | 20.5061  |
| 29      | 4.67  | 1.87  | 101.6  | 1100                       | 723     | 768     | 745.50  | 124.2500                                   | 31.8198  |
| 30      | 4.85  | 1.92  | 103.6  | 1120                       | 672     | 650     | 661.00  | 110.1667                                   | 15.5563  |



**Figure 8:** Plot of average counts per 10s with varying energy of  $\beta$  particles for Sr-22 decay.

## 6 Conclusion

After plotting the two  $\gamma$ -ray spectrum, we see the kinetic energy distribution of the  $\beta$  particles peak at  $E_{\beta+} = 307$  keV and  $E_{\beta-} = 446$  keV for  $^{22}_{11}\text{Na}$  and  $^{90}_{38}\text{Sr}$  respectively.

## References

- Knoll, Glenn F., 2010 *Radiation detection and measurement*. John Wiley & Sons.
- Krane, K.S., 1988. *Introductory nuclear physics*. John Wiley & Sons.